DSA06 — DATA HANDLING AND VISUALIZATION

MAP STORYTELLING CHALLENGE

Slot C • Story Brief MS-C28

High-School Graduates (% of Adults), 1970

with Life Expectancy (1969–71) as the comparison variable

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Activity	Map Storytelling Challenge
Tool	R (ggplot2 + maps)
Marks	20

Story focus
The South region has the lowest mean high-school graduation rate, while the West has the highest. Across all 50 states, the relationship with life expectancy is moderate and positive, but correlation alone does not establish causation.

Prepared as a submission-ready activity report

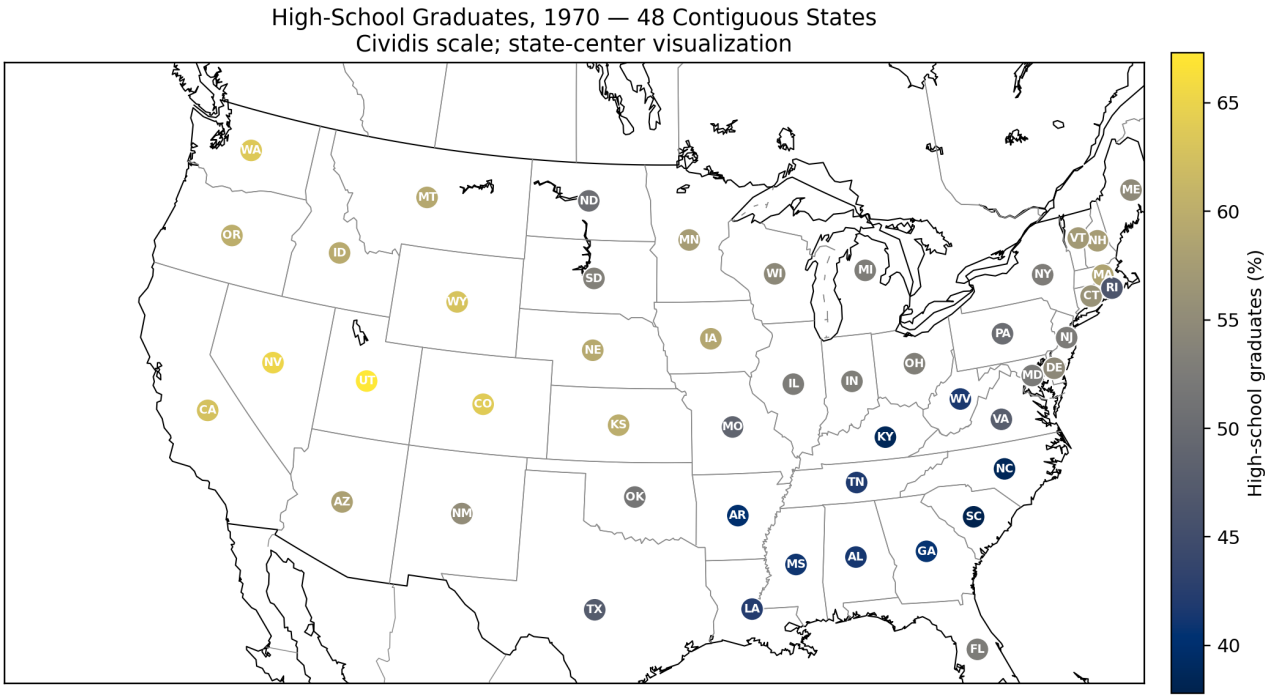
Executive Summary

One clear message
In 1970, the South averaged 44.34% high-school graduates, versus 53.11% across all 50 states. The South therefore reached only 0.83× the national mean, with 1 of 16 Southern states above the national median of 53.25%.

Measure	Result
Highest state	Utah — 67.3%
Lowest state	South Carolina — 37.8%
National median	53.25%
Highest region	West — 62.00%
Lowest region	South — 44.34%
Regional gap	17.66 percentage points
Correlation with life expectancy	r = 0.582 (moderate, positive)

1. The Base Map

The requested map uses a continuous cividis scale and displays the 48 contiguous states. Alaska and Hawaii are excluded from map_data, while all 50 states are retained for statistics.



Note: points are plotted at approximate state centers; the submission code below is the exact ggplot2 + maps polygon map.

Statistic	Answer
Highest HS graduation	Utah — 67.3%
Lowest HS graduation	South Carolina — 37.8%
Median of all 50 states	53.25%

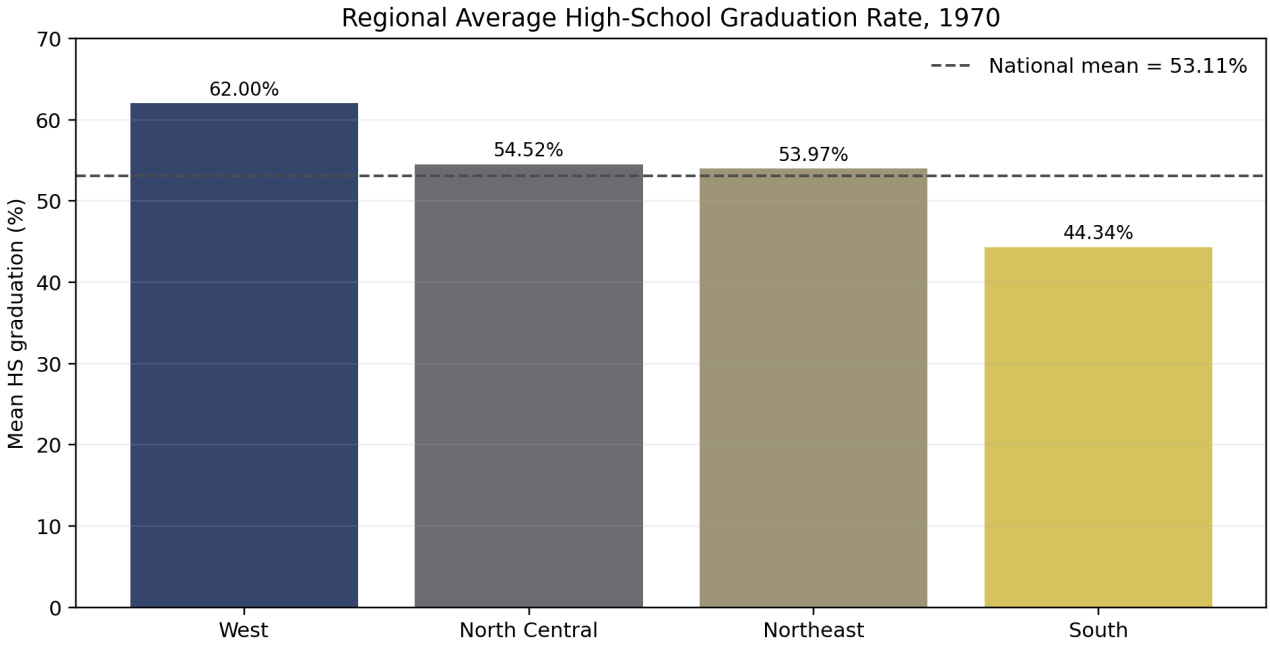
Exact R mapping code requested in the activity:

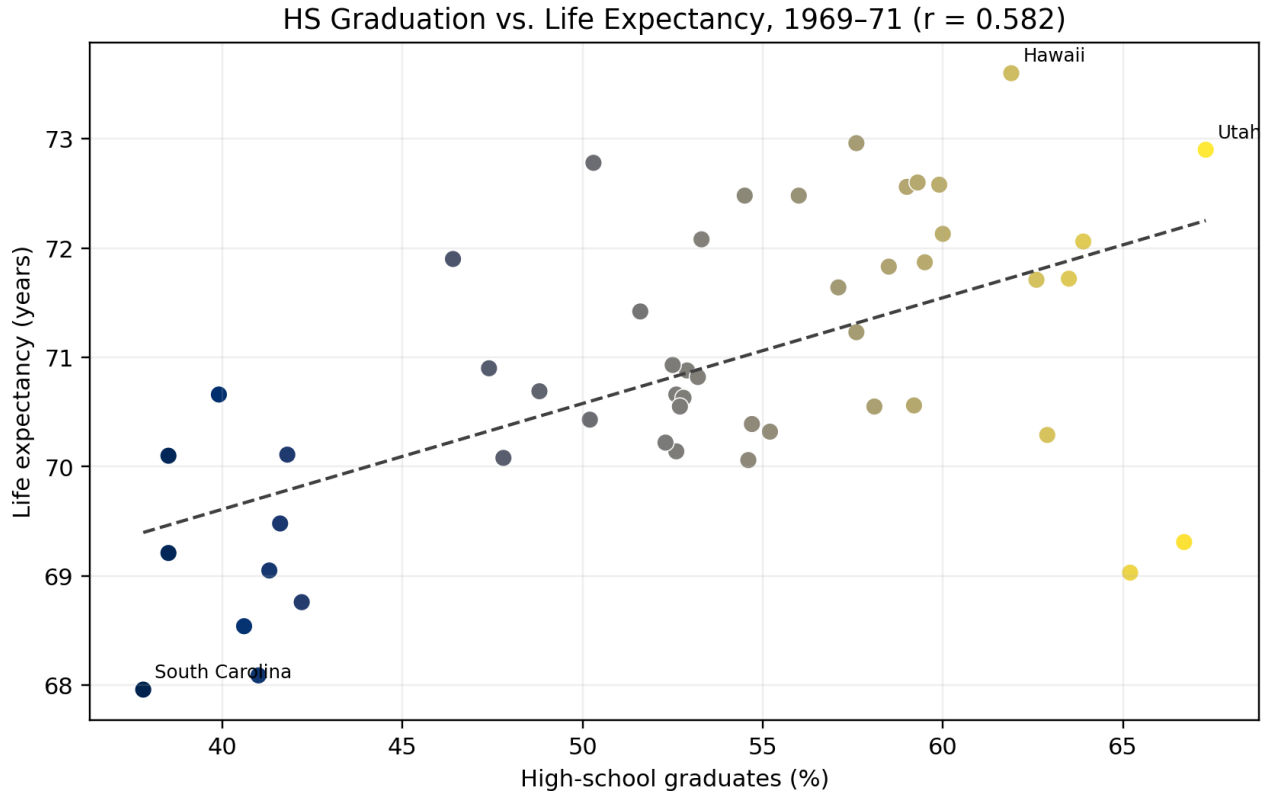
```
ggplot(mapdat, aes(long, lat, group = group, fill = hsgrad)) +
geom_polygon(colour = "white") +
coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
scale_fill_viridis_c(option = "cividis") +
labs(title = "High-School Graduates (% of Adults), 1970",
fill = "HS graduates (%)")
```

2. The Regional Story and Comparison Variable

Region	Mean HS graduates (%)
West	62.00%
North Central	54.52%
Northeast	53.97%
South	44.34%

The West is highest at 62.00%, while the South is lowest at 44.34%; the gap is 17.66 percentage points.





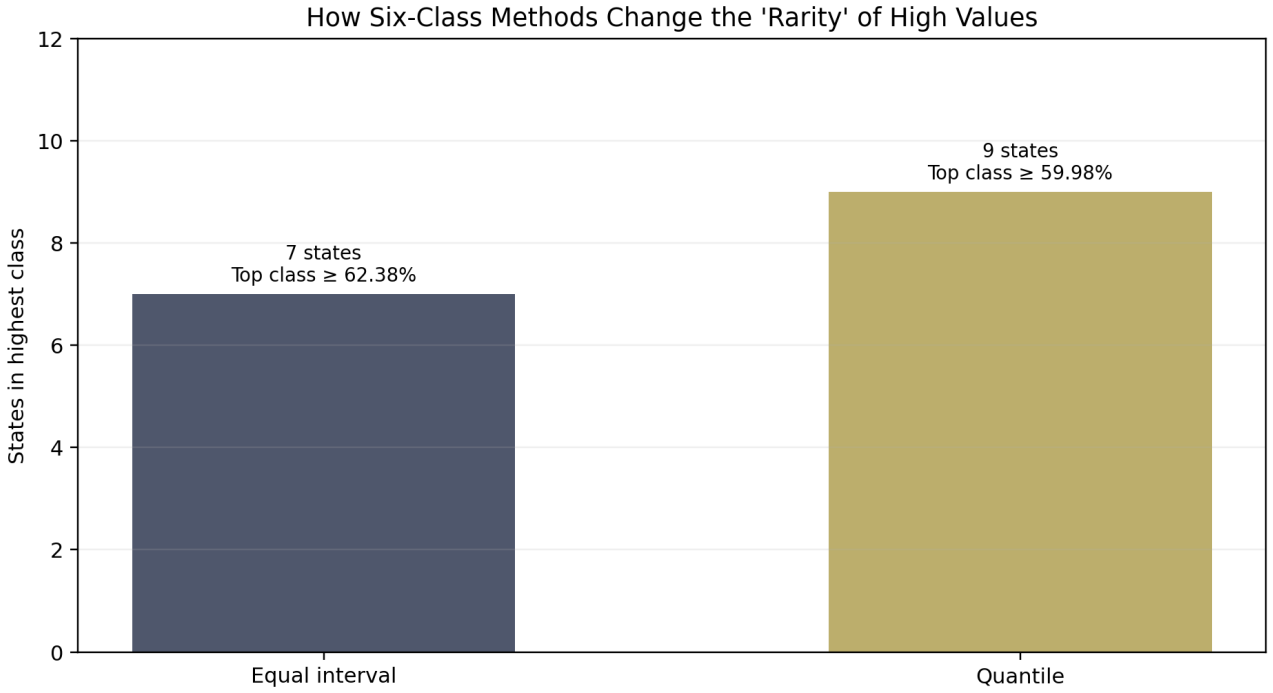
Correlation: r = 0.582. This is a moderate positive association because it lies between 0.3 and 0.7. States with higher high-school graduation rates tend to have higher life expectancy, but this correlation is not proof of cause.

3. Classes Change the Story

Six classes were compared using equal intervals and quantiles. Equal intervals keep the numeric width of every class the same; quantiles keep the number of observations per class approximately balanced.

Method	Top-class lower limit	States in top class
Equal interval	62.38%	7

Method	Top-class lower limit	States in top class
Quantile	59.98%	9



Interpretation: the equal-interval method makes the highest values look rarer because only 7 states enter the top class, compared with 9 under the quantile method. For a local newspaper audience, I would choose quantile classes when the main goal is easy state-to-state comparison without leaving most states in broad middle categories; however, I would clearly label the class method.

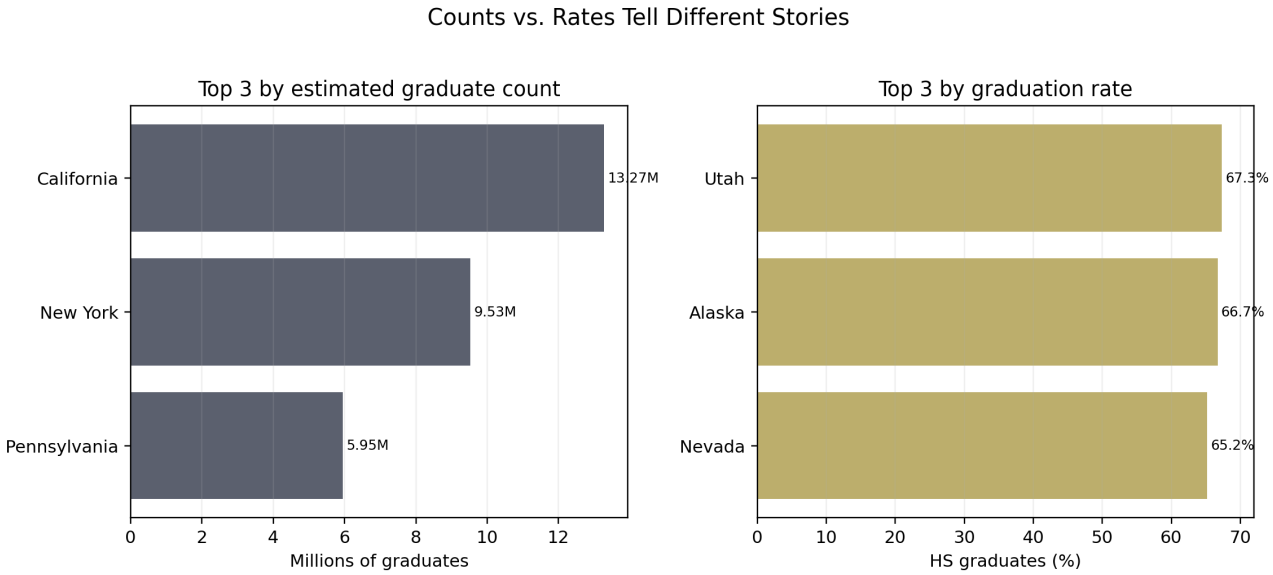
R code used for the two six-class versions:

```
equal_class <- cut(V, 6)
quantile_class <- cut(V, quantile(V, seq(0, 1, length.out = 7)),
include.lowest = TRUE)
```

4. Counts Versus Rates

Estimated number of high-school graduates is calculated as: count = population × 1000 × hsgrad / 100.

Rank	Top by estimated count	Estimated graduates	Top by rate	HS graduation rate
1	California	13,269,948	Utah	67.3%
2	New York	9,526,052	Alaska	66.7%
3	Pennsylvania	5,953,720	Nevada	65.2%



Overlap: 0 states appear in both top-three lists — in fact, none do. A count map would be misleading because large-population states can have many graduates even when their graduation rate is not especially high; the newspaper should map the rate (percentage of adults graduating) rather than the raw count.

5. Telling the Story for the South

South-region measure	Result
Mean HS graduation	44.34%
South / national mean ratio	0.84×
Highest Southern state	Delaware — 54.6%
Southern states above national median	2 of 16

Headline

South trails U.S. in 1970 high-school graduation rates

Evidence and recommendation

The South averaged 44.34% high-school graduates, compared with 53.11% nationally, a ratio of 0.83. Only 2 of the 16 Southern states were above the national median of 53.25%, while Delaware led the region at 54.6%.

Recommendation: publish the South-focused map with the national median and the class method stated beside the legend, so readers can distinguish the regional pattern from the way the data were classified.

R Reproducibility Appendix

The activity supplied the following setup. It is reproduced here without modification so the report can be recreated directly in R.

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area")
d$id <- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$hsgrad
W <- d$life
```

Key calculation commands:

```
max(V); d$state[which.max(V)]
min(V); d$state[V == min(V)]
median(V)
tapply(V, d$zone, mean)
cor(V, W)
```

```
cut(V, 6)
cut(V, quantile(V, seq(0, 1, length.out = 7)), include.lowest = TRUE)
count <- d$pop * 1000 * d$hsgrad / 100
head(d[order(-count), c("state","hsgrad")], 3)
head(d[order(-V), c("state","hsgrad")], 3)
```

Partner Swap Reflection

If I swapped stories with a partner, I would keep the clear South-versus-national comparison but make the legend and class method more prominent. I would also retain the rate-based map rather than a count map, because raw counts mainly reflect population size.

Source Note

Primary dataset: R base package datasets::state.x77 (US State Facts and Figures), containing 50 states and the variables used in this activity. The dataset documentation identifies HS Grad as the percent of high-school graduates in 1970 and Life Exp as life expectancy in years for 1969–71. Source cited in the R documentation: U.S. Department of Commerce, Bureau of the Census (1977), Statistical Abstract of the United States.

Data cross-check: MIT OpenCourseWare's Analytics Edge state-data resource provides a CSV version of the same state dataset. Visuals in this report use a cividis scale to match the activity brief.